

Empathy Map

1. Introduction

Understanding the needs and challenges of end users is an essential part of designing an effective solution. An Empathy Map helps identify what users think, feel, say, and do while interacting with a system. For the OptiCrop project, the primary users are farmers who require accurate crop recommendations based on soil nutrients and environmental conditions. The insights gathered through this empathy map guided the design of a simple, accessible, and data-driven application.

2. Target Users

Primary Users

- Farmers
- Farm Owners

Secondary Users

- Agricultural Officers
- Agronomists
- Agricultural Students
- Researchers

3. Empathy Map

What Users SAY

- "I want to know which crop is best for my land."
- "Weather conditions change every season."
- "I need reliable recommendations before investing."
- "Traditional methods do not always give good results."
- "I prefer simple technology that is easy to use."

What Users THINK

- Will this crop provide better profit?
- Is my soil suitable for cultivation?
- Can technology help improve farming decisions?
- How can I reduce the risk of crop failure?
- I need recommendations based on actual field conditions.

What Users DO

- Collect soil information.
- Consult agricultural experts and experienced farmers.
- Compare different crop options.
- Analyze rainfall and seasonal conditions.
- Use the OptiCrop application to obtain crop recommendations by entering soil and climate parameters.

What Users FEEL

- Concerned about unpredictable farming conditions.
- Uncertain while selecting crops.
- Motivated to improve productivity.
- Confident when recommendations are based on scientific analysis.
- Satisfied when technology helps make better decisions.

4. User Pain Points

Pain Point	Impact
Incorrect crop selection	Reduced agricultural productivity
Lack of scientific guidance	Poor farming decisions
Changing environmental conditions	Increased cultivation risk
Limited access to agricultural experts	Delayed decision making
Traditional farming practices	Lower efficiency and profitability

5. User Needs

The users expect a system that:

- Recommends the most suitable crop accurately.
- Uses scientific analysis instead of assumptions.

- Provides quick prediction results.
- Is simple and user-friendly.
- Supports sustainable farming practices.
- Helps improve agricultural productivity.

6. User Gains

After using OptiCrop, users can expect:

- Better crop selection.
- Increased crop yield.
- Efficient utilization of soil nutrients.
- Reduced financial losses.
- Faster and more informed decision-making.
- Greater confidence in agricultural planning.

7. How OptiCrop Addresses These Needs

The OptiCrop system analyzes seven important agricultural parameters: Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall. These inputs are processed using a trained Logistic Regression model, and the predicted crop is displayed through a simple Flask-based web application, enabling users to make informed cultivation decisions. This directly reflects the implemented prediction workflow in your project.

8. Conclusion

The Empathy Map provides valuable insights into the expectations, challenges, and motivations of farmers. Understanding these user perspectives helped shape the development of OptiCrop as a practical and intelligent crop recommendation system. By combining machine learning with agricultural data, the project supports informed decision-making and contributes to more efficient and sustainable farming practices.